

BicAn: An integrated, open-source framework for polyspectral analysis

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Abstract

We present a novel platform for higher-order spectral analysis of time series data in Python. The theory, utility, and applications of such analyses are summarized. Direct estimation of coherence ($n = 2$), bicoherence ($n = 3$), and tricoherence ($n = 4$) spectra are given for test signals; higher-order ($n > 4$) spectra are inferred at single points in polyfrequency space. Quantification of uncertainty for nonstationary processes is considered, and applications to nonlinear dynamics research are given.

PROGRAM SUMMARY

Program Title: Bicoherence Analyzer (BicAn)

CPC Library link to program files: (to be added by Technical Editor)

Developer's repository link: <https://github.com/rigzridge/pybic>

Code Ocean capsule: (to be added by Technical Editor)

Licensing provisions(please choose one): GPLv3

Programming language: Python

Supplementary material:

Nature of problem:

To detect and characterize nonlinear coupling of various orders given scalar or vector time series data.

Solution method:

Implement a Python module to compute arbitrary-order polyspectra using the short-time Fourier transform (STFT) or continuous wavelet transform (CWT).

Additional comments including restrictions and unusual features:

BicAn supports time-resolved analysis of polyspectral phase coherence, with native 2, 3, and 4D visualization capabilities.

Keywords: Polyspectra; higher-order spectral analysis; nonlinear interaction; phase coherence

1. Introduction to Polyspectra

Despite the well established utility of the Fourier transform to provide information on the spectral content of a digitized signal, the machinery of *higher-order spectra* (HOS) appears to be underrepresented in the typical physicist's toolbox. Indeed, since the Cooley-Tukey fast Fourier transform (FFT) algorithm was published in 1965 [1], the applicability of Fourier analysis has only grown, with the FFT being hailed as the "most important numerical algorithm of the our lifetime" [2]. However, a lesser known¹ work from the same year introduced an extension of Fourier analysis [3]: an infinite hierarchy of spectra based on cumulants.

These generalizations of the familiar power spectrum are known as *polyspectra* or HOS, and have proven useful in various fields. An excellent introduction to HOS, with lucid discussion of applications, is given in [4]. Examples from plasma physics [5, 6, 7, 8, 9, 10, 11, 12] and cosmology [13, 14, 15, 16, 17] are abundant; in industry, HOS are routinely used for fault detection [18, 19, 20]. An overview of the

biomedical utility is found in [21], and consideration of cyclostationary processes is found, *e.g.*, in [22]. Additionally, extensions of select polyspectra to groups have been demonstrated [23], with applications to neural networks and computer vision [24]. An extensive bibliography of 20th century work on HOS may be found in [25].

In this work, we provide a succinct overview of the theory, uses, and implementation of polyspectra. Our intention is neither to be comprehensive in our explanation of application, nor overly rigorous in our presentation of the theory; rather, we seek to furnish an integrated framework for the direct estimation of low-order moment spectra. Furthermore, a need exists for straightforward, portable, and open-source code to conduct these estimations, so that the newly-initiated – as well as those familiar with polyspectral analysis – may have a flexible framework to do so. We posit that BicAn offers just such a utility. Notably, BicAn compliments existing tools (*e.g.*, HOSA [26, 27, 28]) through its visualization capabilities, emphasis on time-localized polyspectra, inclusion of basic signal processing utilities, and support for data exploration via a graphical user interface (GUI).

Crucially, direct application of BicAn to cutting-edge physics problems has been demonstrated [29]. In short, the PyBic mod-

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¹Approximately 700 citations compared to almost 20000 for [1].

ule (which implements BicAn in Python) was used to identify time-resolved polyspectral “signatures” of nonlinear coupling between unstable eigenmodes in the DIII-D tokamak [30]. Because enhanced amplitude of these eigenmodes may facilitate the loss of energetic particles in magnetically-confined fusion experiments [31, 32], advances in the detection and characterization of such nonlinearity aid not only the understanding, but also the mitigation, of such deleterious particle transport. Thus, while the most immediate applications of BicAn are found in the fusion community, we believe this software offers a benefit to scientists across myriad fields and subdisciplines.

For brevity, we define HOS in terms of central moment functions instead of cumulants, which is somewhat unorthodox, but not unprecedented [33, 34]. Though a cumulant-based approach has desirable properties for stochastic signals, a framework based on moments is conceptually simpler, and is well-founded for the analysis of deterministic signals [35]. In fact, the n th-order central moments are equivalent to the n th-order cumulants for $n \leq 3$, and in general, cumulants may be completely reconstructed from knowledge of lower-order moments [3, 5, 36].

The paper is outlined as follows: Section 2 elucidates the mathematical underpinnings of our approach, while Section 3 tackles the task of implementing these formulae in Python. Section 4 is concerned with benchmarking the code with a battery of test signals, and Section 5 discusses applications and future work.

2. Mathematical foundation

2.1. Generalized correlation

Given a real-valued, integrable function $x : \mathbb{R} \rightarrow \mathbb{R}$ of a single variable (t , say), consider the correlation of $x(t)$ with $n-1$ independently translated copies $x(t + \tau_i)$ [3, 4, 35, 36, 37, 38],

$$R_x(\tau_1, \dots, \tau_{n-1}) = \langle x(t + \tau_1) \dots x(t + \tau_{n-1}) x(t) \rangle, \quad (1)$$

where $\langle \cdot \rangle$ is an expectation operator (w.r.t. t), and $\{\tau_i\}_{i=1}^{n-1}$ represent a set of delays or *lags*. Likewise, if given a set of $n > 1$ real-valued, integrable functions of a single variable $\{x_i\}_{i=1}^n$, we consider the more general moment function,

$$R_{x_1 \dots x_n}(\tau_1, \dots, \tau_{n-1}) = \langle x_1(t + \tau_1) \dots x_{n-1}(t + \tau_{n-1}) x_n(t) \rangle. \quad (2)$$

We call (2) the n th-order moment function, or the n th-order correlation, of the functions $\{x_i\}_{i=1}^n$. Clearly, when $x_i \rightarrow x$, we recover $R_{x_1 \dots x_n} \rightarrow R_x$. If each of the x_i have compact support and a vanishing mean ($\langle x_i \rangle = 0$), it is conventional to write

$$\begin{aligned} R_{x_1 \dots x_n}(\tau_1, \dots, \tau_{n-1}) &= \int_{-\infty}^{\infty} dt x_1(t + \tau_1) \dots x_{n-1}(t + \tau_{n-1}) x_n(t), \\ &= \int_{-\infty}^{\infty} dt x_n(t) \prod_{i=1}^{n-1} x_i(t + \tau_i). \end{aligned} \quad (3)$$

Recalling the unitary Fourier transform and its inverse,

$$\hat{x}(f) \equiv \int_{-\infty}^{\infty} dt x(t) e^{-2\pi i f t} = \mathcal{F}\{x(t)\}(f), \quad (4a)$$

$$x(t) = \int_{-\infty}^{\infty} df \hat{x}(f) e^{2\pi i f t} = \mathcal{F}^{-1}\{\hat{x}(f)\}(t), \quad (4b)$$

we express (3) as

$$R_{x_1 \dots x_n}(\tau_1, \dots, \tau_{n-1}) = \int_{-\infty}^{\infty} dt x_n(t) \prod_{i=1}^{n-1} \int_{-\infty}^{\infty} df_i \hat{x}_i(f_i) e^{2\pi i f_i(t + \tau_i)}. \quad (5)$$

Upon interchange of integrations, we find that $R_{x_1 \dots x_n}$ is given by an $(n-1)$ -dimensional *inverse* Fourier transform; hence, the Fourier transform of the n th-order moment function is given by

$$\widetilde{S}_{x_1 \dots x_n} \equiv \widehat{R}_{x_1 \dots x_n}(f_1, \dots, f_{n-1}) = \overline{\hat{x}_n\left(\sum_{j=1}^{n-1} f_j\right)} \prod_{i=1}^{n-1} \hat{x}_i(f_i) \in \mathbb{C}, \quad (6)$$

noticing $\hat{x}(-f) = \overline{\hat{x}(f)}$ for real-valued functions, where an overbar implies complex conjugation $a + ib \equiv a - ib$. When estimating (6) for digitized data, it is customary to ensemble average over a number of realizations in time. Thus, in practice, our estimation of the n th-order polyspectrum $\mathcal{S}_{x_1 \dots x_n}$ (*i.e.*, the n th-order moment spectrum, or n -spectrum) is found via

$$\mathcal{S}_{x_1 \dots x_n}(f_1, \dots, f_{n-1}) = \frac{1}{M} \sum_{j=1}^M \widetilde{S}_{x_1 \dots x_n}^{(j)} = \left\langle \overline{\hat{x}_n\left(\sum_{j=1}^{n-1} f_j\right)} \prod_{i=1}^{n-1} \hat{x}_i(f_i) \right\rangle, \quad (7)$$

where $\widetilde{S}_{x_1 \dots x_n}^{(j)}$ is the n th-order moment spectrum for the j th realization, and we have assumed uniform weighting over M realizations.

Written this way, we see that the n th-order moment spectrum quantifies phase-coherency between n fluctuations with frequencies satisfying $f_1 + \dots + f_{n-1} = f_n$. Note that when the amplitudes $|\widetilde{S}_{x_1 \dots x_n}^{(j)}|$ are constant, (7) becomes

$$\mathcal{S}_{x_1 \dots x_n}(f_1, \dots, f_{n-1}) = \langle |\widetilde{S}_{x_1 \dots x_n}^{(j)}| e^{i\varsigma(f_1, \dots, f_{n-1})} \rangle \propto \langle e^{i\varsigma(f_1, \dots, f_{n-1})} \rangle, \quad (8)$$

where the n -phase ς is given by

$$\varsigma(f_1, \dots, f_{n-1}) = \left(\sum_{i=1}^{n-1} \varphi_i(f_i) \right) - \varphi_n\left(\sum_{j=1}^{n-1} f_j\right), \quad (9)$$

defining the phases φ_i via $\hat{x}_i(f) = |\hat{x}_i(f)| e^{i\varphi_i(f)}$. Hence, for a stationary signal, the n -spectrum $\mathcal{S}_{x_1 \dots x_n}$ will tend to vanish whenever the phasor $e^{i\varsigma}$ is randomly distributed in time. Conversely, preference for a certain angle is indicative of consistent phase relationships between the n fluctuations, *i.e.*, phase-coherency. We discuss this in more detail in §4.2.

While this sensitivity to simultaneous phase dynamics is the hallmark of polyspectral analysis, the n th-order moment spectrum does not compensate for differences in amplitude of the participating fluctuations. Thus, normalizing (7) with the Cauchy-Schwartz inequality [5] yields the (squared) n th-order coherence spectrum,

$$s_{x_1 \dots x_n}^2(f_1, \dots, f_{n-1}) = \frac{|\mathcal{S}_{x_1 \dots x_n}(f_1, \dots, f_{n-1})|^2}{\left\langle \left| \prod_{i=1}^{n-1} \hat{x}_i(f_i) \right|^2 \right\rangle \left\langle \left| \hat{x}_n\left(\sum_{j=1}^{n-1} f_j\right) \right|^2 \right\rangle} \in [0, 1]. \quad (10)$$

In principle, this measures the proportion of energy within process x_n , at frequency $f_n = f_1 + \dots + f_{n-1}$, that derives from phase coherence with fluctuations $\hat{x}_1(f_1), \dots, \hat{x}_{n-1}(f_{n-1})$ [5]. In the special case of a single process $x_i \rightarrow x$, we call $\mathcal{S}_x \equiv \mathcal{S}_{x\dots x}$ or $s_x^2 \equiv s_{x\dots x}^2$ the n th-order *auto*-polyspectrum or *auto*-coherence spectrum, respectively. By inspection of (7), we note that auto-polyspectra satisfy many symmetry relations, *e.g.*, the arguments $f_1, \dots, f_{n-1}$ may be freely permuted. Thus, when considering auto-polyspectra, we limit our interest to the *principal domain* $f_1 > f_2 > \dots > f_{n-1} > 0$, subject to the Nyquist constraint $f_1 + \dots + f_{n-1} \leq f_{\text{Nyq}}$ (explained in the next section).

For clarity, suppose $n = 2$. Then the second-order correlation is the familiar *cross-correlation*,²

$$R_{x_1 x_2}(\tau) = \int_{-\infty}^{\infty} dt x_1(t + \tau) x_2(t), \quad (11)$$

and the second-order moment spectrum is the so-called *cross-spectrum*,

$$\mathcal{C}_{x_1 x_2}(f) \equiv \mathcal{S}_{x_1 x_2}(f) = \langle \hat{x}_1(f) \overline{\hat{x}_2(f)} \rangle. \quad (12)$$

Note that when $x_1 = x_2 = x$, we recover the usual power spectrum, $\mathcal{S}_{xx}(f) = |\hat{x}(f)|^2$. Normalizing the cross-spectrum yields the squared *cross-coherence spectrum*,

$$c_{x_1 x_2}^2(f) \equiv s_{x_1 x_2}^2(f) = \frac{|\mathcal{C}_{x_1 x_2}(f)|^2}{\langle |\hat{x}_1(f)|^2 \rangle \langle |\hat{x}_2(f)|^2 \rangle}. \quad (13)$$

Setting $n = 3$ gives the *cross-bispectrum*,

$$\mathcal{B}_{x_1 x_2 x_3}(f_1, f_2) \equiv \mathcal{S}_{x_1 x_2 x_3}(f_1, f_2) = \langle \hat{x}_1(f_1) \hat{x}_2(f_2) \overline{\hat{x}_3(f_1 + f_2)} \rangle, \quad (14)$$

and its normalized counterpart, the squared *cross-bicoherence spectrum*:

$$b_{x_1 x_2 x_3}^2(f_1, f_2) \equiv s_{x_1 x_2 x_3}^2(f_1, f_2) = \frac{|\mathcal{B}_{x_1 x_2 x_3}(f_1, f_2)|^2}{\langle |\hat{x}_1(f_1) \hat{x}_2(f_2)|^2 \rangle \langle |\hat{x}_3(f_1 + f_2)|^2 \rangle}. \quad (15)$$

Similarly, $n = 4$ provides the *cross-trispectrum*,

$$\begin{aligned} \mathcal{T}_{x_1 x_2 x_3 x_4}(f_1, f_2, f_3) &\equiv \mathcal{S}_{x_1 x_2 x_3 x_4}(f_1, f_2, f_3) \\ &= \langle \hat{x}_1(f_1) \hat{x}_2(f_2) \hat{x}_3(f_3) \overline{\hat{x}_4(f_1 + f_2 + f_3)} \rangle, \end{aligned} \quad (16)$$

and the squared *cross-tricoherence spectrum*,

$$\begin{aligned} t_{x_1 x_2 x_3 x_4}^2(f_1, f_2, f_3) &\equiv s_{x_1 x_2 x_3 x_4}^2(f_1, f_2, f_3) \\ &= \frac{|\mathcal{T}_{x_1 x_2 x_3 x_4}(f_1, f_2, f_3)|^2}{\langle |\hat{x}_1(f_1) \hat{x}_2(f_2) \hat{x}_3(f_3)|^2 \rangle \langle |\hat{x}_4(f_1 + f_2 + f_3)|^2 \rangle}, \end{aligned} \quad (17)$$

where this use of t^2 should not be confused with our symbol for time. For $n > 4$, the form of the n th-order coherence spectrum is obvious.

²Note that two conventions exist for the cross-correlation. Here, we follow that given by (3), which is consistent with, *e.g.*, [39]. However, the roles of x_1 and x_2 tend to be exchanged in mathematics textbooks, such as [40].

2.2. Time-frequency representation

Fundamental to our approach is the time-frequency representation (TFR) of a signal. Though any complex-valued TFR with a linear frequency scale is acceptable in principle, we focus on the short-time Fourier transform (STFT) and continuous wavelet transform (CWT). The STFT is simply the Fourier transform of a windowed signal [41],

$$\text{STFT}_x(f, \tau) \equiv \mathcal{F}\{x(t)w(t - \tau)\}(f, \tau) = \int_{-\infty}^{\infty} dt x(t)w(t - \tau)e^{-2\pi i f t}, \quad (18)$$

where $w(\cdot) \in L^2$ is a square-integrable window function (*i.e.*, $\int_{-\infty}^{\infty} |w(t)|^2 dt < \infty$). Similarly, the CWT is given by [6]

$$\text{CWT}_x(\alpha, \tau) \equiv \alpha^q \int_{-\infty}^{\infty} dt x(t) \overline{\Psi_\sigma(\alpha(t - \tau))}, \quad (19)$$

using the unitless parameter α and the time-like parameter τ to scale and shift t , respectively. The exponent q may be freely chosen, but typically [42] takes values in the range $q \in [0, 1]$. The function $\Psi_\sigma(t)$ is the titular wavelet; to connect with the conventions of Fourier analysis, we choose the *Morlet* function [43],

$$\Psi_\sigma(\alpha t) = \pi^{-1/4} \sqrt{\frac{\alpha}{\sigma}} \exp\left[-\frac{1}{2} \left(\frac{\alpha t}{\sigma}\right)^2 + 2\pi i \alpha f_0 t\right], \quad (20)$$

which is simply a complex exponential windowed by a Gaussian envelope. The time-like parameter σ governs the extent of the wavelet in the time domain, and enables fine control of the simultaneous time and frequency resolution. The “base” frequency f_0 is arbitrary. Note that the prefactor in (20) guarantees $\int_{-\infty}^{\infty} |\Psi_\sigma(\alpha t)|^2 dt = 1$, *i.e.*, the wavelet is normalized with respect to the L^2 norm.

We shall see in the next section that when applied to discrete data, the STFT is directly estimated through an analogue of (18); conversely, the CWT is most efficiently estimated in frequency space via an inverse Fourier transform,

$$\text{CWT}_x(\alpha, \tau) = \pi^{1/4} \sqrt{2\sigma} \alpha^{q-1/2} \mathcal{F}^{-1} \left\{ \hat{x}(f) e^{-2\pi^2 \sigma^2 \left(\frac{f}{\alpha} - f_0\right)^2} \right\}(\tau). \quad (21)$$

Note that this function is centered where $f = \alpha f_0$, and so our scale relates linearly to frequency. Remarkably, we find that choosing $q = 1/2$ eliminates all dependence on α in the prefactor.

3. Implementation

3.1. Code overview

BICAN (*Bicoherence Analyzer*) is an object-oriented framework for the direct estimation and exploration of low-order moment spectra, with the capability to investigate polyspectra of arbitrary order. The **BICAN** object is invoked using the **PyBic** module; module functions supply a lightweight signal-processing toolbox, in addition to general-purpose utilities, *e.g.*,

`PlotLabels()` simplifies the labeling of figures. (Unless otherwise noted, typewriter font indicates an attribute, such as `SampRate`, or a function/method, like `TestSignal()`.)

Instances of a `BicAn` object conveniently subsume these lower-level functions within class methods, while holding relevant parameters and data as attributes. Thus, many class methods are wrapper functions, handling input of parameters for the STFT or CWT, say, rather than compelling the user to deal with these details manually. This dependency on module functions enables users to painlessly augment their analyses without rewriting the source code, or misusing existing class methods. Simply stated, this structure enforces a code which is both flexible and extensible. As an example, Fig. 1 illustrates the sequence of method calls used in the `BicAn.ProcessData()` class method. Crucially, the code is intended to be as self-

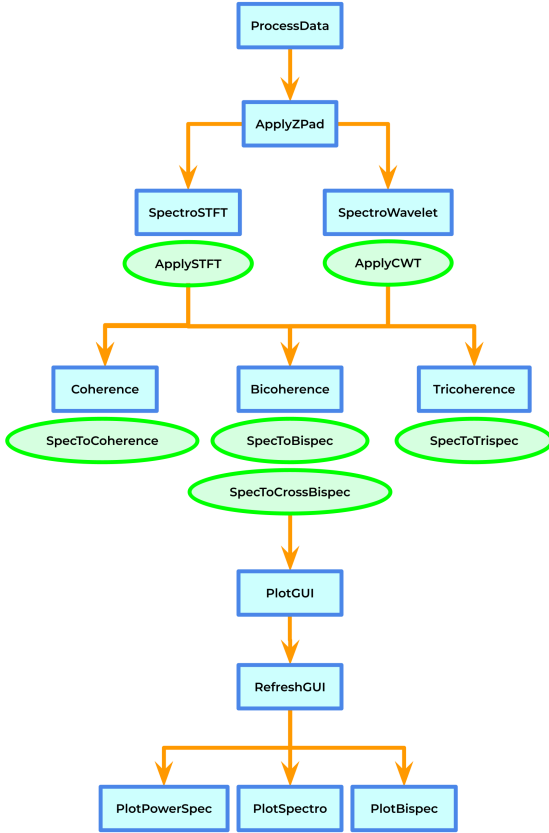


Figure 1: Flowchart of method calls in execution of `BicAn.ProcessData()` class method. Blue boxes represent class methods, while green ellipses represent module functions.

contained as possible: dependency is limited to the NumPy [44], SciPy [45], Matplotlib [46], and Tkinter [47] packages, which are included in many – if not most – Python distributions. A more comprehensive understanding of the class methods, class attributes, and module functions is available from the source code on [GitHub](#). Note that two web-hosted Jupyter notebooks (also available on [GitHub](#)) have been developed to accompany this publication: one provides a guided tour of `PyBic`, and another reproduces the figures herein (see §5). A dedicated user guide *per se* is in revision.

The current state of the software was driven by a need to improve both versatility and throughput. Code development began with a relatively inflexible, purpose-built MATLAB GUI [48] that assessed bicoherence in magnetics data from the DIII-D tokamak; it is the spiritual successor to a program written in the mid-1990s [49]. Further enhancements incentivized the adoption of an object-oriented paradigm, from which `BicAn` was born.

3.2. Digitization

To estimate the n th-order moment spectrum for discrete data, we first compute a TFR by applying the results of §2.2 to one or more time series. For our purposes, a time series implies a sequence of N real-valued data gathered at regularly spaced intervals in time, at a sampling rate f_s . That is,

$$x \rightarrow \{x[t_j]\}_{j=0}^{N-1} \Rightarrow x(t) \rightarrow x[j], \quad (22)$$

where $t_j = j\Delta t$, using $\Delta t = 1/f_s$. (Note that we use square brackets instead of parentheses to indicate the piecewise nature of x .) In this case, the Fourier transform becomes the discrete Fourier transform (DFT),

$$\hat{x}(f) \rightarrow \hat{x}[k] \equiv \frac{1}{N} \sum_{j=0}^{N-1} x[t_j] \exp[-2\pi i f_k t_j], \quad (23)$$

where $f_k = k/N\Delta t$.

To obtain a discrete TFR $X[j, k]$, `BicAn` calculates the STFT or CWT with the `ApplySTFT()` or `ApplyCWT()` module functions, respectively. For an instance of a `BicAn` object, these are consequently wrapped by the `SpectroSTFT()` and `SpectroWavelet()` class methods. Explicitly, our code uses the discrete version of (18) for the STFT,

$$X_n^{\text{STFT}}[j, k] = \frac{1}{L} \sum_{m=0}^{L-1} x_n[m + j\lambda] w[m] \exp\left[-\frac{2\pi i m k}{L}\right], \quad (24)$$

where $L \leq N$ samples is the subinterval size (`BicAn.SubInt`), stepped by λ samples (`BicAn.Step`); $w[\cdot]$ is a window L samples wide (`BicAn.Window`). By default, `BicAn` uses the Hann function [50], $w[m] = \sin^2[m\pi/(L-1)]$, though other windows such as the flat-top [51] are supported. Note that if σ is redefined such that $\sigma \rightarrow \sigma N\Delta t = \sigma N/f_s$, discretizing the CWT (21) yields the convenient form,

$$X_n^{\text{CWT}}[j, k] = \mathcal{A}_\sigma k^{q-1/2} \sum_{p=0}^{N-1} \hat{x}_n[p] \exp\left[-\frac{2\pi^2 \sigma^2}{k^2} (p-k)^2 + \frac{2\pi i j p}{N}\right], \quad (25)$$

with constant prefactor $\mathcal{A}_\sigma \equiv \pi^{1/4} \sqrt{2N\sigma/f_s}$, where the values of σ and q are governed by the `BicAn.Sigma` and `BicAn.AlphaExp` attributes, respectively. Note that our calculation of either $X_n^{\text{STFT}}[j, k]$ or $X_n^{\text{CWT}}[j, k]$ leverages the FFT algorithm for efficiency.

In practice, the choice of TFR is indicated by the user with the `BicAn.SpecType` attribute, and hence

$$X_n[j, k] = \begin{cases} X_n^{\text{STFT}}[j, k], & \text{SpecType='stft'}, \\ X_n^{\text{CWT}}[j, k], & \text{SpecType='cwt'}, \end{cases} \quad (26)$$

As written, the STFT splits time series into subintervals, while the CWT uses contiguous data. Thus, in BicAn, the TFR has different dimensions, namely $M_{\text{STFT}} \times \lfloor L/2 \rfloor$ for the STFT (where $M_{\text{STFT}} = 1 + \lfloor (N - L)/\lambda \rfloor$), and $\lfloor N/2 \rfloor \times \lfloor N/2 \rfloor$ for the CWT. Note that the Nyquist frequency $f_{\text{Nyq}} \equiv f_s/2$ is located at index $k_{\text{Nyq}} = \lfloor L/2 \rfloor$ or $k_{\text{Nyq}} = \lfloor N/2 \rfloor$ for the STFT or CWT, respectively. Since $L \leq N$, this implies a difference in frequency resolution, with minimum frequencies $f_{\text{min}}^{\text{STFT}} = f_s/L$ or $f_{\text{min}}^{\text{CWT}} = f_s/N$. Note that the nature of the CWT enforces an effective frequency resolution which varies with frequency, *e.g.*, high frequencies are distributed across many Fourier bins [52].

With this, we define the spectrogram as the square modulus of a given TFR $X[j, k]$,

$$\text{spectrogram}[j, k] = \tilde{S}_{xx}[j, k] = |X[j, k]|^2. \quad (27)$$

Likewise, the power spectrum $\mathcal{P}$ is the second-order moment spectrum, an ensemble average of a given TFR over all realizations,

$$\mathcal{P}_x[k] = S_{xx}[k] = \langle |X[j, k]|^2 \rangle = \frac{1}{M} \sum_{j=0}^{M-1} |X[j, k]|^2. \quad (28)$$

The discrete n th-order moment spectrum is then estimated for M realizations with

$$S_{x_1 \dots x_n}[k_1, \dots, k_{n-1}] = \frac{1}{M} \sum_{j=0}^{M-1} \tilde{S}_{x_1 \dots x_n}[j, k_1, \dots, k_{n-1}], \quad (29)$$

subject to the Nyquist restriction $k_1 + \dots + k_{n-1} \leq k_{\text{Nyq}}$, and the local [53] (*i.e.*, “instantaneous” [48]) n -spectrum is given by

$$\tilde{S}_{x_1 \dots x_n}[j, k_1, \dots, k_{n-1}] = X_1[j, k_1] \dots X_{n-1}[j, k_{n-1}] \overline{X_n[j, k_n]}, \quad (30)$$

where $k_n \equiv k_1 + \dots + k_{n-1}$. The discrete n th-order coherence spectrum takes the form

$$s_{x_1 \dots x_{n-1}}^2[k_1, \dots, k_{n-1}] = \frac{|S_{x_1 \dots x_n}[k_1, \dots, k_{n-1}]|^2}{E_{x_1 \dots x_{n-1}}[k_1, \dots, k_{n-1}] E_{x_n}[k_1, \dots, k_{n-1}] + \varepsilon}, \quad (31)$$

where the small number $\varepsilon \ll 1$ prevents $0/0 = \text{NaN}$ errors (controlled by BicAn.Epsilon), and the normalization is given by

$$E_{x_1 \dots x_{n-1}}[k_1, \dots, k_{n-1}] = \frac{1}{M} \sum_{j=0}^{M-1} |X_1[j, k_1] \dots X_{n-1}[j, k_{n-1}]|^2, \\ E_{x_n}[k_1, \dots, k_{n-1}] = \frac{1}{M} \sum_{j=0}^{M-1} |X_n[j, k_1 + \dots + k_{n-1}]|^2. \quad (32)$$

BicAn supplies the GetPolySpec() method for direct calculation of (29) & (31). For more common or specific cases, class methods use TFR data to estimate low-order polyspectra in whole or in part; an example usage of Coherence(), Bicoherence(), and Tricoherence() is considered in the next section.

4. Analysis of Test Signals

4.1. Stationary modes

Because the n th-order *auto*-coherence spectrum is trivial for $n < 3$ (note that $s_{xx}^2(f) = 1$ by (10)), the most straightforward test of the code is also the simplest diagnostic case, *i.e.*, the auto-bispectrum $\mathcal{B}_x(f_1, f_2)$ of the sum of three sinusoids:

$$x(t) = \cos(2\pi f_a t + \phi) + \cos(2\pi f_b t) + \cos(2\pi f_c t) + \text{noise}(t), \quad (33)$$

where $\phi = \phi(t)$ is a (possibly) time-dependent phase, $f_c = f_a + f_b$, and the last term represents additive white noise $\in [-1, 1]$. Aside from this noise, the frequency content of $x(t)$ does not vary with time, thus $d|\mathcal{B}_x|/dt = 0$ to an excellent approximation. If $f_a > f_b$ and $\phi(t) = 0$, we expect a single peak in the principal domain at $\mathcal{B}_x(f_a, f_b)$ and $b_x^2(f_a, f_b)$.

For convenience, BicAn is equipped with a number of test signals, all of which may be easily accessed using the TestSignal() module function. In particular, '3tone' implements (33) with $\phi = 0$, $f_a = 45$ Hz and $f_b = 19$ Hz; sampling is performed at $f_s = 200$ Hz. As predicted, BicAn reports a lone peak near $(f_1, f_2) = (45, 19)$ Hz (see Fig. 2).

Similarly, considering four static modes,

$$x(t) = \cos(2\pi f_a t) + \cos(2\pi f_b t) + \cos(2\pi f_c t) + \cos(2\pi f_d t) + \text{noise}(t), \quad (34)$$

where now $f_d = f_a + f_b + f_c$, we expect the auto-trispectrum to have a single peak (in the principal domain) at $\mathcal{T}_x(f_a, f_b, f_c)$. Choosing $(f_a, f_b, f_c) = (40, 32, 12)$ Hz ('4tone'), we clearly observe this peak in the auto-tricoherence spectrum $t_x^2(f_a, f_b, f_c)$ in Fig. 3. More generally, in the case of n static modes, BicAn provides MonteCarloMax(). This class method employs a simple Monte-Carlo hillclimb (*i.e.*, a random-restart hillclimb [54]) to identify maxima in high-dimensional manifolds. *Forthcoming versions will include a power spectrum-weighted random number distribution.*

4.2. Dynamic phase and/or modulus

Amplitude and/or phase modulation of the local n -spectrum fundamentally affects its ensemble average, *i.e.*, $S_{x_1 \dots x_n}(f_1, \dots, f_{n-1})$. Mathematically, this is given by the first equality in (8). If, for a given point in poly-frequency space $(f_1, \dots, f_{n-1})$, the n -phase is modeled by

$$\zeta(f_1, \dots, f_{n-1}, t) = \zeta_0 + \omega_0 t + \theta \sin \omega t, \quad (35)$$

then whenever $\theta \neq 0$, the phasor average $\langle e^{i\zeta} \rangle$ over a time interval $T = 2\pi/\omega$ becomes exactly

$$\langle e^{i\zeta(t)} \rangle = \frac{1}{T} \int_0^T e^{i(\zeta_0 + \omega_0 t + \theta \sin \omega t)} dt = e^{i(\zeta_0 + \eta\pi)} J_\eta(\theta), \quad (36)$$

where $\eta \equiv \omega_0/\omega$, and $J_\eta(\theta)$ is a Bessel function of the first kind, of order η . Figure 4 illustrates this correspondence for $\eta = 0$.

When $\theta = 0$, we see that $\langle e^{i\zeta} \rangle \propto \sin(\pi\eta)/\pi\eta$, proving our previous claim that for a stationary time series, the n th order polyspectrum tends to vanish where the n -phase varies randomly, or linearly, for a sufficiently long interval ($T \propto \eta/\omega_0$). Note that if $\theta = 0$, (36) vanishes identically for all $\eta \in \mathbb{N}/0$.

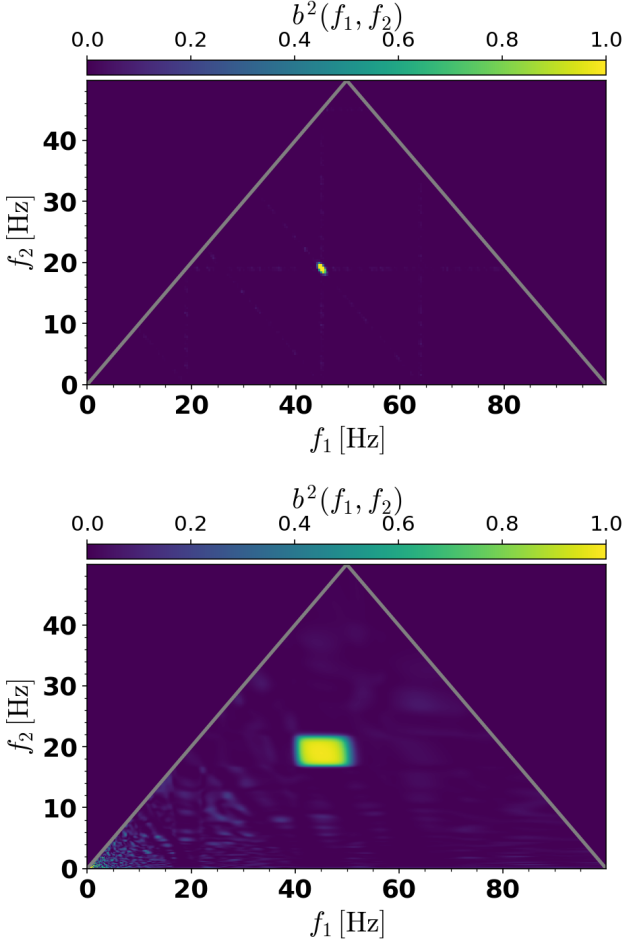


Figure 2: Auto-bicoherence spectra $b_x^2(f_1, f_2)$ of diagnostic time series (33) using STFT for $t \in [0, 100]$ s [top] and CWT for $t \in [0, 10]$ s [bottom], as implemented by `Bicoherence()` class method. STFT uses $L = 512$ sample subinterval, $\lambda = 128$ sample step, and Hann window; CWT uses $\sigma = \pi$ and $q = 1/2$. Both plots produced with `PlotBispec()`; gray triangles delineate principal domain. Note the differences in apparent frequency resolution.

For a generic, time-dependent local polyspectrum (of any order) $\tilde{S}(t)$, we consider the average over an arbitrary time interval T ,

$$\mathcal{S} = \langle \tilde{S}(t) \rangle = \frac{1}{T} \int_0^T |\tilde{S}(t)| e^{i\mathcal{S}(t)} dt, \quad (37)$$

where we have dropped subscripts and frequency dependencies for clarity. In this form, we observe that when T is sufficiently large, the dominant contributions to the integral will occur when the n -phase is stationary $\dot{\zeta}(t) \approx 0$, *i.e.*, during phase coherence. Thus, the n th-order polyspectrum is most sensitive to intervals of coincident n -phase stationarity and enhanced local polyspectral modulus.

To demonstrate this principle with synthetic data, we again consider the sum of three sinusoids (33), but now allow for a linear phase $\phi(t) = \omega_0 t$; as before, $(f_a, f_b) = (45, 19)$ Hz ('linear_phase'). In the absence of noise, the phase of the local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$ (*i.e.*, the 3-phase, or *biphase*

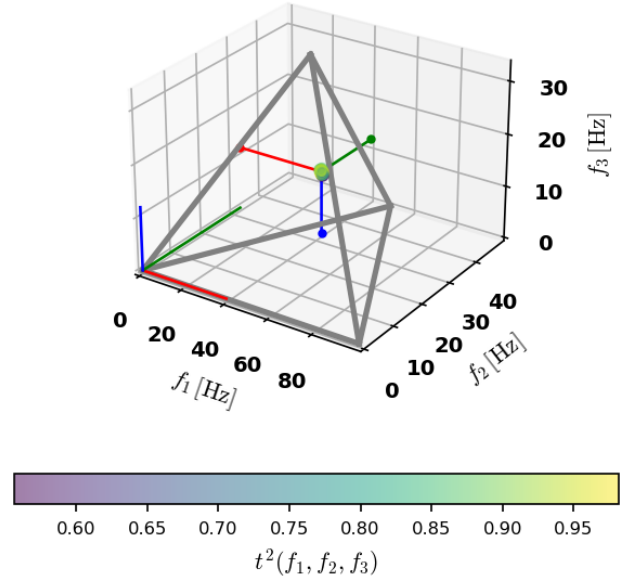


Figure 3: Scatterplot of auto-tricoherence spectrum of diagnostic time series (34) over interval $t \in [0, 100]$ s, using Hann-windowed STFT ($L = 512$, $\lambda = 128$), with `Tricoherence()` and `PlotTrispec()` class methods. Threshold for plotting is $t_x^2(f_1, f_2, f_3) = 50\%$. Interior of gray tetrahedron represents principal domain; red, green, and blue lines lie along the f_1 , f_2 , and f_3 axes, respectively, and assist in locating the peak value of tricoherence ($t^2 \approx 98.6\%$) in 3D space.

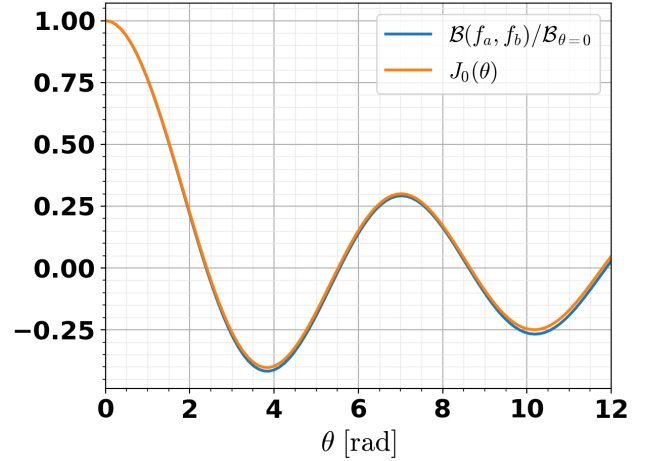


Figure 4: Comparison of Bessel function $J_0(\theta)$ with normalized 3-spectrum (bispectrum). To make this figure, we estimate the bispectral value $\mathcal{B}_x(f_a, f_b)$ of (33) for $\phi(t) = \theta \sin \omega t$, using $\omega/2\pi = 10$ mHz and $\theta \in [0, 12]$ rad. Calculation performed by `GetBispec()` method.

$\beta(f_a, f_b, t)$ also satisfies $\beta(f_a, f_b, t) = \omega_0 t$, and thus both the bispectrum $\mathcal{B}_x(f_a, f_b)$ and bicoherence spectrum $b_x^2(f_a, f_b)$ will tend to zero. Indeed, Fig. 5 presents the evolution of the (normalized) local bispectrum as inferred by `BICAN`, which clearly follows the unit circle.

Crucially, we note that a calculation of the complete bispectrum $\mathcal{B}_x(f_1, f_2)$ is not required to produce such a figure.

The `PlotPointOut()` method (which, in this case, wraps `GetBispec()`), enables direct access to the time-resolved bispectrum at an arbitrary point in bi-frequency space (f_1, f_2), where the graphical output of the method is determined by the `WhatPlot` attribute. For example, `PlotPointOut()` can be configured to return a statistical analysis, as shown in Fig. 6. Such statistical analyses utilize the noise floor and standard deviation given by [34, 52, 55, 56], in conjunction with the methodology of [53, 57], *i.e.*, we estimate a probability density function (PDF) by randomizing the TFR phase and recalculating the n th-order coherence spectrum many times N_{trials} . From this distribution (which tends to be well-modeled by an exponential [55, 58]), a bicoherence threshold representing confidence at a desired level may be deduced. Note that in the case of a linear biphas, Fig. 6 reports the expected null bicoherence, $b_x^2(f_a, f_b) \approx 0.03\%$. For completeness, we include the probability distribution of biphas across time (see Fig. 6 inset), which reflects the uniform phase angle coverage from Fig. 5.

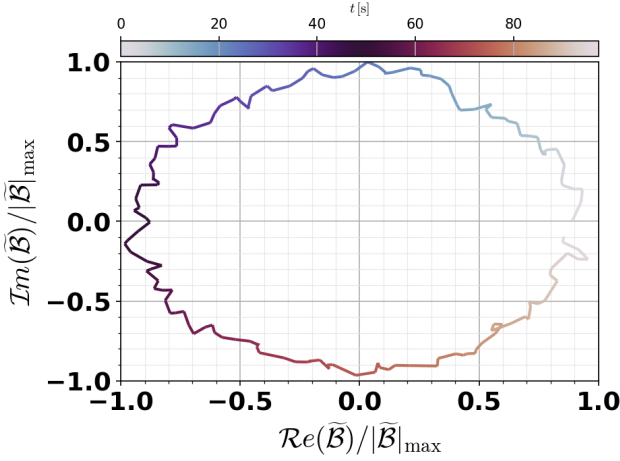


Figure 5: Time evolution of local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$ (normalized to maximum modulus $|\tilde{\mathcal{B}}|_{\max}$) for diagnostic time series (33), where $\phi(t) = \omega_0 t$, with $\omega_0/2\pi = 10$ mHz. Figure given by `PlotPointOut()` class method with `WhatPlot='hybrid'`, color represents time evolution.

Conversely, Figs. 7 & 8 demonstrate that a given polyspectrum does not necessarily vanish for a modulated n -phase. Specifically, we consider (33) with $\phi(t) = \omega_0 t + (\pi/2) \sin \omega t$ and $(f_a, f_b) = (45, 19)$ Hz ('phase_mod'). Note that the difference between the observed values of $b_x^2(f_a, f_b)$ (compare Figs. 6 & 8) is wholly due to $\phi(t)$. Unlike the evenly distributed phasors in Fig. 5, accumulation of the phase dynamics reported by Fig. 7 will yield preference for a given angle (in this case, π radians). In addition to the aforementioned methods of uncertainty quantification, a bootstrap technique [59] is employed to estimate the PDF for the observed value of $b_x^2(f_a, f_b)$. Explicitly, we calculate the bispectrum and squared bicoherence spectrum from a randomly resampled (with replacement) local bispectrum $\tilde{\mathcal{B}}_x[j, k_1, k_2]$. Repeating this process many times ($N_{\text{trials}} \geq 1000$) provides a robust PDF which requires no underlying assumptions of Gaussianity; an example is given in Fig. 8.

Naturally, the interpretation of polyspectra is more compli-

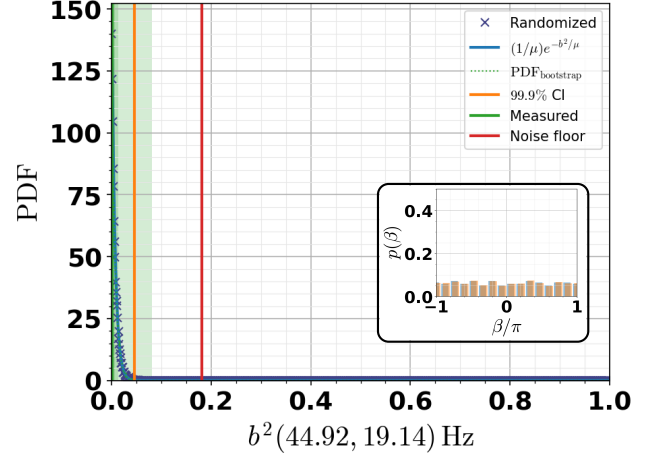


Figure 6: Probability density function (PDF) of single value of bicoherence spectrum $b_x^2(f_a, f_b)$ using $N_{\text{trials}} = 10000$ phase-randomized realizations, as given by `PlotPointOut()` class method with `WhatPlot='bicoch'`. Blue curve gives exponential fit to binned data (blue $\times$ s), orange line shows 99.9% confidence level, and red line is estimated noise floor. Green line represents measured b^2 , and light green shaded regions report both the sample standard deviation of the randomized data and the bicoherence uncertainty $1/\sqrt{M}$ [52, 58]. Inset reports the measured distribution of biphas using an unweighted [blue] and local bispectral modulus-weighted [orange] histogram.

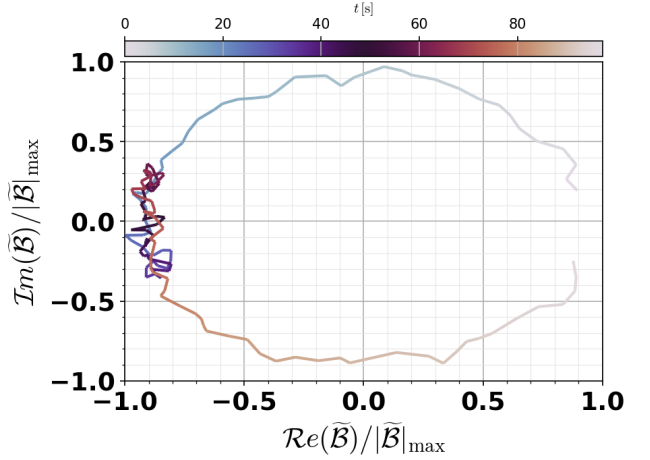


Figure 7: Time evolution of normalized local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$ for diagnostic time series (33), where $\phi(t) = \omega_0 t + (1/2) \sin \omega t$, with $\omega_0/2\pi = \omega/2\pi = 10$ mHz. Note that an ensemble average of these approximate phasors will yield preference for π radians.

cated if the polyspectral modulus is allowed to vary in time, as phase-incoherence may accumulate in the presence of highly transient, large-amplitude fluctuations. In the most pathological case, the values assigned to a short burst of samples so far exceed the magnitude of their cohort that the time series is effectively noise besides these handful of data. (Indeed, this would invalidate our working assumption of uniform weighting $\langle x \rangle = \sum_{i=1}^N x[j]/N$.) More realistically, the modulus of a given Fourier coefficient, calculated from noisy data, will tend

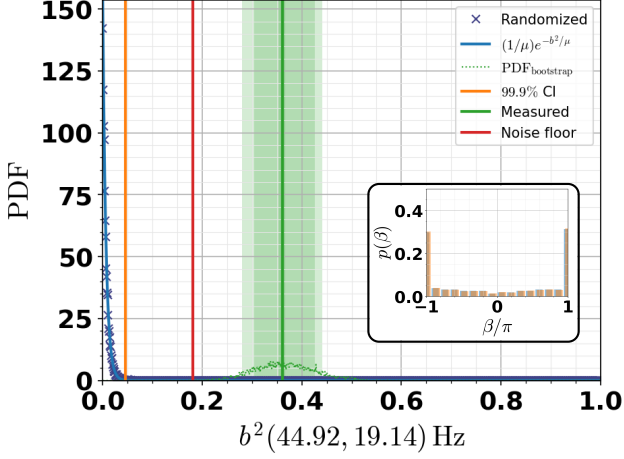


Figure 8: Statistical assessment for single value of squared bicoherence spectrum $b_x^2(f_a, f_b)$ for case of linear biphas plus phase modulation. Bootstrap PDF given by dotted green line; overlapping green regions report, in decreasing magnitude, $1/\sqrt{M}$, the bootstrap standard deviation, and the sample standard deviation from phase randomization. Note the preference for $\beta \approx \pi$ radians (inset), and the statistically significant value of bicoherence $\approx 36\%$.

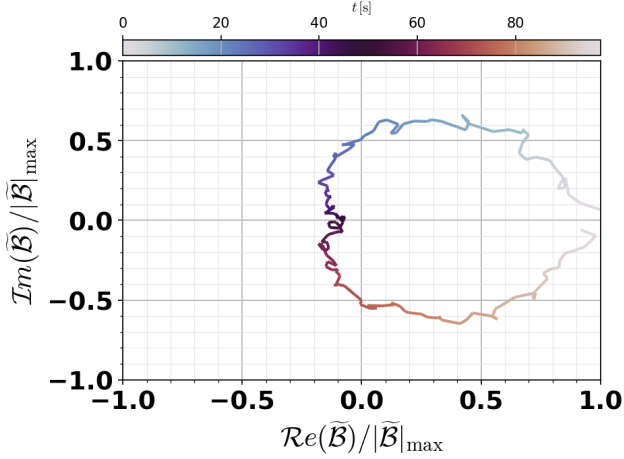


Figure 9: Time evolution of normalized local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$ for amplitude-modulated time series with linear phase (38).

to vary over time, and hence the quantification of polyspectral uncertainty is critical to a complete analysis.

As an example, we discretize a generalization of (33),

$$x(t) = \cos(2\pi f_a t + \phi(t)) + \cos(2\pi f_b t) + A(t) \cos(2\pi f_c t) + \text{noise}(t), \quad (38)$$

where the phase of the first term and amplitude of the third term are given by $\phi(t) = \omega_0 t$ and $A(t) = 1 - 0.9 \sin^2(\omega_0 t/2)$, respectively, with $\omega_0/2\pi = 10$ mHz and $(f_a, f_b) = (45, 19)$ Hz ('linear_phase_am'). In Fig. 9, the amplitude of the local bispectrum $|\tilde{\mathcal{B}}_x(f_a, f_b)|$ is seen to modulate as the biphas advances linearly $\beta(f_a, f_b) \propto t$. For this case, BicAn estimates a value of bicoherence ($b^2 \approx 11\%$) below the noise floor ($b_{\text{noise}}^2 \approx 18\%$), but above the 99.9% confidence level inferred by our randomized phase procedure ($b_{99.9\%}^2 \approx 5\%$), as shown

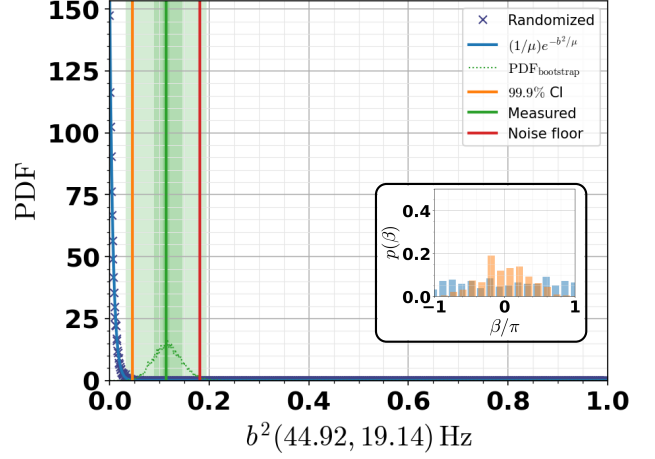


Figure 10: Statistical assessment for single value of squared bicoherence spectrum $b_x^2(f_a, f_b)$ for case of linear biphas but modulated local bispectral modulus. Note that despite the bulk of the bootstrap PDF residing below the noise floor, it does tend to exceed the 99.9% confidence level for a randomized biphas. Crucially, differences in unweighted [blue] and local bispectral modulus-weighted [orange] probabilities concur with the local bispectrum distribution shown in Fig. 9.

in Fig. 10. Thus, in practice, it is important to calculate and consider both the theoretical and inferred noise floors, as only the latter is materially dependent on any given data. In fact, although the identification of phase coherence in Fig. 10 is statistically inconclusive, the coordinated phase and amplitude dynamics seen in Fig. 9 may justify further analysis if observed in real data, especially if such coincidence persists over multiple cycles.

To briefly investigate simultaneous modulation of the local polyspectral modulus and n -phase, we retain the form of (38), now with amplitude $A(t) = 1 - 0.9 \cos^2(\omega_0 t/2)$, and consider a modulated phase $\phi(t) = \omega_0 t - \pi/2 + \sin \omega_0 t$, using $\omega_0/2\pi = 10$ mHz and $(f_a, f_b) = (45, 19)$ Hz ('phase_mod_am'). Explicitly, this signal exhibits a biphas which becomes stationary $\dot{\beta} \rightarrow \dot{\phi} = \omega_0(1 + \cos \omega_0 t) \approx 0$ where $\omega_0 t \approx \pi$, precisely when the local bispectral modulus is maximized. Figure 11 visualizes the evolution of the corresponding local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$. Crucially, the dynamics in Fig. 11 are similar to local bispectral signatures generated by a time-dependent nonlinearity [29], and are therefore associated with significant phase coherence. Figure 12 presents the statistical analysis of $b_x^2(f_a, f_b)$, which estimates bicoherence in excess of 50%.

To better highlight the coincidence of an enhanced local polyspectral modulus and stationary n -phase in indicating transient phase coherence, we consider the *instantaneous difference frequency*,

$$\Delta f_{\text{inst}}(t) \equiv \frac{1}{2\pi} \frac{d\zeta(t)}{dt}. \quad (39)$$

The `PlotInstFreq()` class method has been developed to visualize the local bispectral modulus vs. instantaneous difference frequency (as calculated from biphas); Fig. 13 reveals that, for the data in Fig. 11, enhanced $|\tilde{\mathcal{B}}_x(f_a, f_b)|$ is indeed associated with $\dot{\beta} \approx 0$.

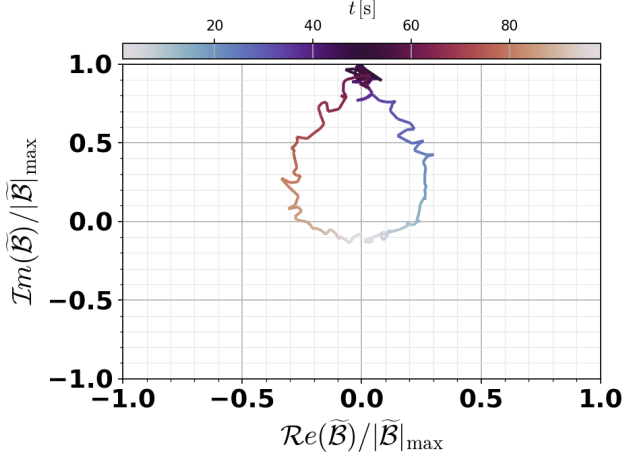


Figure 11: Time evolution of normalized local bispectrum $\tilde{\mathcal{B}}_x(f_a, f_b)$ during simultaneous modulation of the local polyspectral modulus and n -phase; approximately stationary biphase is observed near $t = 50$ s.

4.3. Cross-polyspectra

BicAN conveniently supports the direct estimation of second and third-order cross-polyspectra in the entire principal domain. For higher orders of cross-polyspectra, `GetPolySpec()` allows estimations at single points in poly-frequency space *i.e.*, $(f_1, \dots, f_{n-1})$ for arbitrary n . Figure 14 compares the power spectrum and second-order coherence spectrum $s_{x_1 x_2}^2$ (*i.e.*, the cross-coherence spectrum $c_{x_1 x_2}^2$) for the signals

$$\begin{aligned} x_1(t) &= \cos(2\pi f_a t) + \cos(2\pi f_b t) + \text{noise}(t), \\ x_2(t) &= \cos(2\pi f_b t) + \text{noise}(t), \end{aligned} \quad (40)$$

where we retain $(f_a, f_b) = (45, 19)$ Hz ('coherence'). Note that significant coherence ~ 1 is observed only at the shared frequency f_b , as expected.

A more involved example is given by the set of frequency-modulated (FM) signals:

$$\begin{aligned} x_1(t) &= \sin(2\pi f_a t + \Delta_a), \\ x_2(t) &= \sin(2\pi f_b t + \Delta_b), \\ x_3(t) &= \sin(2\pi f_c t + \Delta_c), \end{aligned} \quad (41)$$

where $f_c = f_a + f_b$ and $\Delta_c = \Delta_a + \Delta_b$, using

$$\begin{aligned} \Delta_a(t) &= \frac{A_f}{f_\Delta} \sin(2\pi f_\Delta t), \\ \Delta_b(t) &= \frac{A_f}{f_\Delta} \cos(2\pi f_\Delta t), \end{aligned} \quad (42)$$

with amplitude A_f and modulation frequency f_Δ . Choosing $(f_a, f_b, f_\Delta, A_f) = (45, 19, 0.05, 10)$ Hz ('cross_circle'), we present the cross-bicoherence spectrum $b_{x_1 x_2 x_3}^2(f_1, f_2)$ as estimated by BicAN in Fig. 15. Each "circle" observed in this figure is due to the accumulation of a single, time-dependent feature over the course of the interval. Note that this is conceptually distinct from the static (albeit modulated) tones we have thus considered, and will be briefly explored in the next subsection.

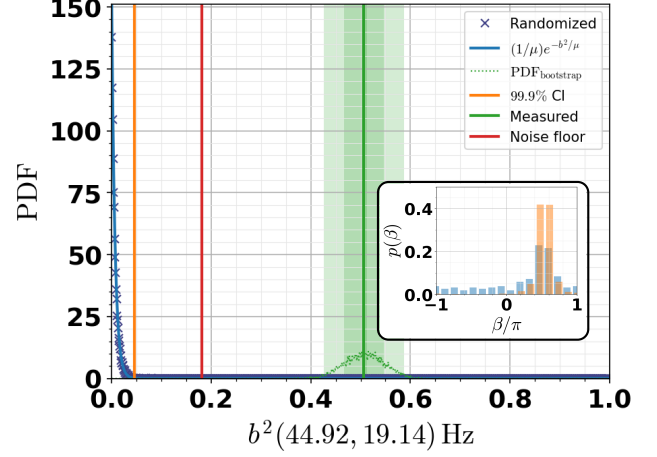


Figure 12: Statistical assessment for single value of squared bicoherence spectrum $b_x^2(f_a, f_b)$ for modulated biphasic and varying local bispectral modulus. Note the highly statistically significant value of bicoherence $\approx 51\%$, as well as the peak in *unweighted* biphase probability.

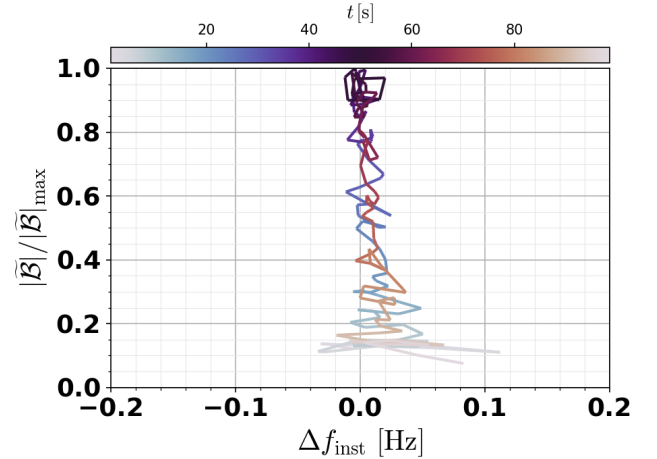


Figure 13: Instantaneous difference frequency $\Delta f_{\text{inst}} (= \dot{\beta}/2\pi)$ vs. local bispectral modulus. Crucially, a large local modulus is correlated with stationary biphase, *i.e.*, $\Delta f_{\text{inst}} \approx 0$.

4.4. Time evolution in poly-frequency space

Generally speaking, the frequency content of a time series tends to vary with time. If, by some process (adiabatic change of equilibrium quantities, say), a solitary, statistically significant polyspectral feature slowly follows a "path" through poly-frequency space during an interval of time, then two observations may be made. First, at any given time, the local n -spectrum $\mathcal{S}_{x_1 \dots x_n}$ will contain a single polyspectral feature (by definition). Next, assuming the path does not cross itself, the total n th-order coherence spectrum will tend to weight the points along the path equally [48]. As discussed, circular features in Fig. 15 derive from this mechanism.

To emphasize this phenomenon, we use BicAN to construct the TFR, power spectrum, and auto-bicoherence spectrum as

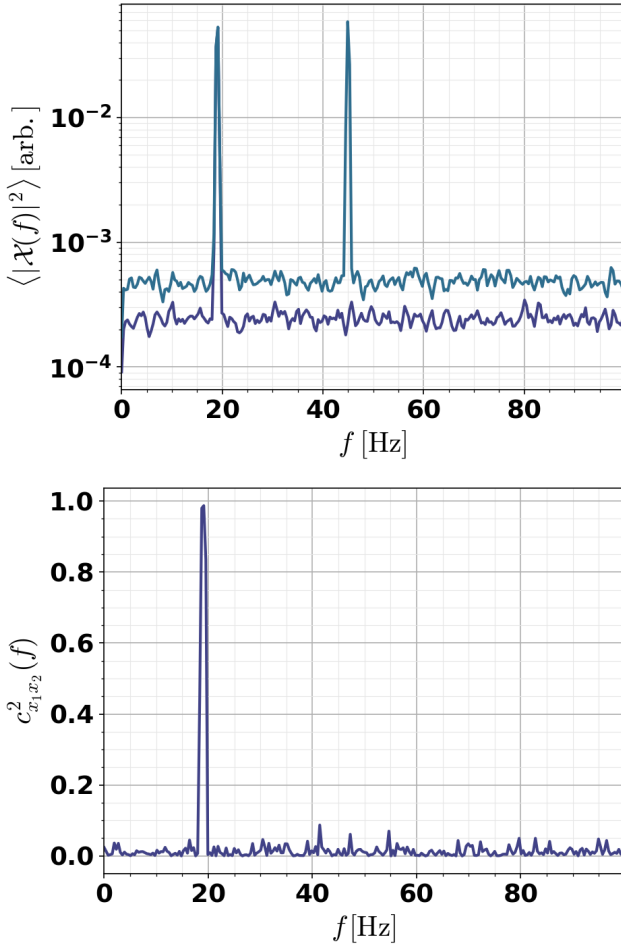


Figure 14: Power spectra [top] and cross-coherence spectrum [bottom] of two time series containing a shared tone (40), calculated with `Coherence()` class method. Visualizations produced by `PlotPowerSpec()` and `PlotCoherence()` class methods, respectively. Note lone coherent peak in lower panel.

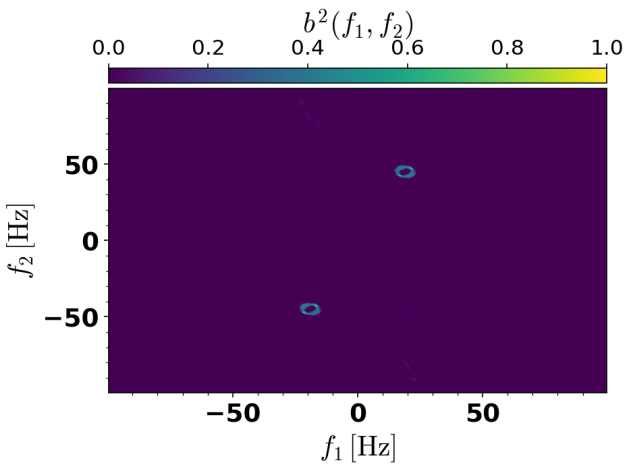


Figure 15: Squared cross-bicoherence spectrum $b^2_{x_1 x_2 x_3}(f_1, f_2)$ estimated from set of diagnostic time series (41) with additive white noise. Underlying TFR is STFT ($L = 512$, $\lambda = 128$, flat-top window). Note appearance of conjugate features near $(f_1, f_2) = (f_a, f_b)$, $(-f_a, -f_b)$.

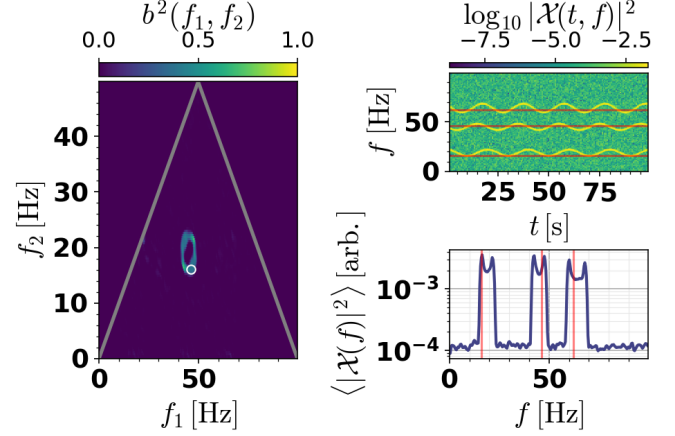


Figure 16: Reproduction of `PlotGUI()` output for sum of three FM tones (41). Note that auto-bicoherence spectrum [left], spectrogram [upper right] and power spectrum [lower right] are produced by `PlotBispec()`, `PlotSpectro()`, and `PlotPowerSpec()` class methods, respectively. STFT uses $L = 512$, $\lambda = 128$ and flat-top window. Red lines in right plots emulate user feedback from clicking the bicoherence spectrum near $(f_1, f_2) = (46, 16)$ Hz [white circle].

sociated with the time series

$$x(t) = x_1(t) + x_2(t) + x_3(t) + \text{noise}(t), \quad (43)$$

with $x_i(t)$ as given by (41) ('circle'). We consider the auto-bicoherence for its visual simplicity, and because `BicAn` facilitates the direct exploration of data for third-order polyspectra with the `PlotGUI()` class method (see Fig. 16). Most usefully, this method provides an interactive figure wherein clicks to the bicoherence spectrum translate to responsive, user-guided application of `PlotPointOut()`, as illustrated in Fig. 17. A freely-available Jupyter notebook (<https://tinyurl.com/BicAnTutorial>) gives a comprehensive, guided tour of `PlotGUI()`'s capabilities.

Figure 18 presents the sensitivity of `BicAn` in characterizing a time-dependent bispectral feature at a single point in bi-frequency space. On the one hand, we observe that the mean bicoherence estimate (green line) does not vary appreciably for any considered combination of parameters, *i.e.*, `BicAn` provides a consistent estimation of $b^2(f_a, f_b)$. On the other hand, we find that for a fixed subinterval L , the bootstrap standard deviation and 99.9% confidence level are enhanced for larger step size λ ; this enhancement is seen to increase with L . Recalling the discussion of §3.2, we find that reducing the step or subinterval size yields greater STFT time resolution, and hence provides a larger statistical ensemble to average (albeit at the expense of frequency resolution in the latter case). In contrast, when emulating this analysis with a CWT, the 99.9% confidence level was only very weakly correlated with the σ parameter in the range $\sigma \in [1, 8.5]$.

As a final example, we generate a slowly-varying fourth-order polyspectral feature using a generalization of (34) & (41),

$$x(t) = \sum_{i \in \{a, b, c, d\}} \sin(2\pi f_i t + \Delta_i + \phi_i) + \text{noise}(t), \quad (44)$$

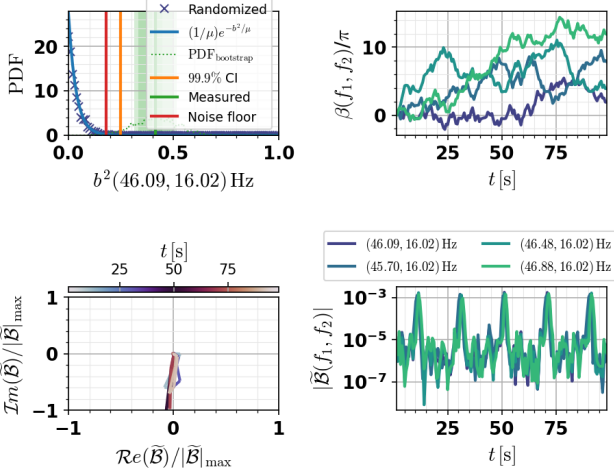


Figure 17: Time-resolved analysis of single point in bi-frequency space $(f_1, f_2) \approx (46, 16)$ Hz, as generated by right-clicking bicoherence spectrum in PlotGUI() window (see Fig. 16). Clockwise from top left, we see uncertainty quantification ($N_{\text{trials}} = 200$), biphase/ π , local bispectral modulus, and normalized local bispectrum. Note that each individual figure may also be produced interactively via hotkeys and a left-click.

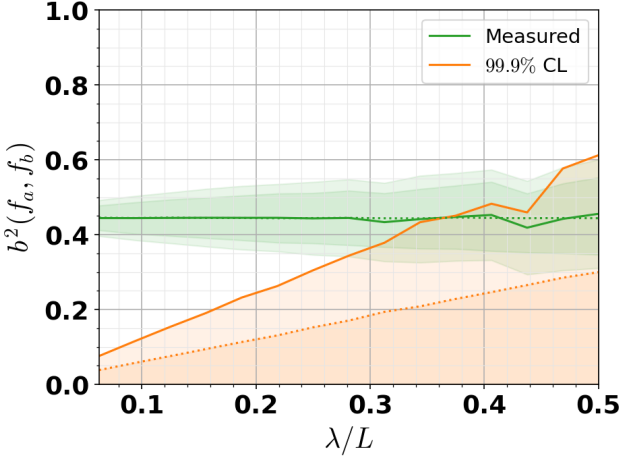


Figure 18: Sensitivity analysis of bootstrap bicoherence estimate [green] and 99.9% confidence level [orange] vs. step/subinterval ratio λ/L for the 'circle' test signal (43), at $(f_1, f_2) \approx (46, 16)$ Hz. Solid and dotted lines indicate $L = 512$ and $L = 256$ samples, respectively.

where $f_d = f_a + f_b + f_c$ and $\Delta_d = \Delta_a + \Delta_b + \Delta_c$, with

$$\begin{aligned}\Delta_a(t) &= \frac{A_f(t)}{f_\Delta} \sin(2\pi f_\Delta t), \\ \Delta_b(t) &= \frac{A_f(t)}{f_\Delta} \cos(2\pi f_\Delta t), \\ \Delta_c(t) &= 2\pi \times 10(t/T)^2.\end{aligned}\quad (45)$$

The amplitude of frequency modulation decays exponentially as $A_f(t) = A_{f0}e^{-20t/T}$, and the phases are chosen such that $\phi_a(t) = \omega_0 t$, and $\phi_b = \phi_c = \phi_d = 0$; i.e., the 4-phase (triphas) of the trispectral feature will vary linearly. Setting $T = 10$ s

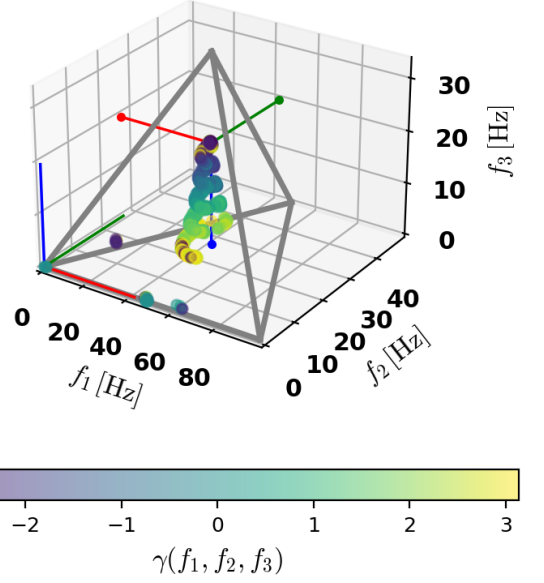


Figure 19: Auto-tricoherence spectrum of 'helix' test signal (44). Figure created with PlotTrispec(), using $t_x^2(f_1, f_2, f_3) = 15\%$ threshold; color is determined by triphase $\gamma(f_1, f_2, f_3)$.

and $(f_a, f_b, f_c, f_\Delta, A_{f0}) = (42, 25, 0, 0.05, 10)$ Hz, BicAN reports a tapering helix of tricoherence whose triphase changes continuously ('helix', see Fig. 19).

5. Discussion

We have presented a flexible, open-source toolkit for polyspectral analyses of arbitrary order. Polyspectra have been defined as multidimensional Fourier transforms of n th-order moments, and their digital implementation has been articulated. Analysis of example cases have affirmed the developed theory and oriented insight on practical issues for interpreting polyspectral analyses, e.g., calculation of confidence levels. The presence of modulation in the local polyspectral modulus and/or n -phase at a single point in poly-frequency space was shown to play a key role in the resulting n -spectrum and n -coherence spectrum. Additionally, cross-polyspectral analysis was demonstrated, and generic paths through poly-frequency space have been investigated.

We note that throughout this work, we have considered only confidence levels for single points in poly-frequency space; however, to construct confidence levels for large swaths of poly-frequency space (e.g., bicoherence maps), the false discovery rate (FDR) should be minimized by implementing, e.g., the Benjamini and Hochberg procedure [60], as done in [53]. Such features will be included in future versions of the code. Another prospective avenue of improvement is the strengthening of BicAN's preprocessing abilities: while standard techniques such as zero padding (ApplyZPad()), detrending (ApplyDetrend()), bandpass filtering (ApplyBandpass()),

and interpolation (`WhittakerShannon()`) are presently supported, inclusion of, *e.g.*, multitaper [61] methods or a prewhitening [62] option would be advantageous for cross-polyspectral analyses of disparate data, where differences in preprocessing contribute to spurious spectral features. (Note that the current version accepts only time series of uniform length and sampling rate.)

As alluded to in the introduction, the potential applications of `BicAn` are numerous. Besides its basal utility as a signal-processing tool, `BicAn` is adept at identifying the coupling between multiple fluctuations, *e.g.*, as observed in a quadratic or cubic nonlinearity. In the former case, we consider a generic time series of the form

$$\begin{aligned} x(t) &= \sum_i \cos(2\pi f_i t) + \sum_{i>j} \cos(2\pi f_i t) \cos(2\pi f_j t), \\ &= \cdots + \frac{1}{2} \sum_{i>j} \left\{ \cos[2\pi(f_i + f_j)t] + \cos[2\pi(f_i - f_j)t] \right\}, \end{aligned} \quad (46)$$

where the indices i, j run over some arbitrary number of modes, and we have expanded the nonlinear term with a trigonometric identity in the second line. Note that due to the generation of modes at the sum and difference frequencies, a polyspectral analysis will report statistically significant bicoherence at the points $(f_1, f_2) = (f_i, f_j)$ and $(f_1, f_2) = (f_j, f_i - f_j)$. Such analyses are common in the literature.

Plasma physicists, in particular, have long been associated with bispectral analysis for this purpose [5], hence the recent use of `BicAn` to detect transient quadratic coupling between magnetic fluctuations in the DIII-D tokamak [29]. A forthcoming publication identifies “instantaneous difference frequency locking” ($\Delta f_{\text{inst}} \approx 0$) of eigenmodes during these brief periods ($\sim 100 \mu\text{s}$) of nonlinear interaction [63].

Funding

Partial funding from DoE Office of Science grant DE-SC0021404 is gratefully acknowledged.

Data Availability Statement

The data and figures herein may be viewed or reproduced via <https://tinyurl.com/BicAnPlots>.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J. W. Cooley, J. W. Tukey, An algorithm for the machine calculation of complex fourier series, *Mathematics of computation* 19 (90) (1965) 297–301, <https://doi.org/10.1090%2FS0025-5718-1965-0178586-1>.
- [2] G. Strang, Wavelets, *American Scientist* 82 (3) (1994) 250–255, <https://www.jstor.org/stable/29775194>.
- [3] D. R. Brillinger, An introduction to polyspectra, *The Annals of Mathematical Statistics* (1965) 1351–1374 <https://www.jstor.org/stable/2238424>.
- [4] C. L. Nikias, M. R. Raghuveer, Bispectrum estimation: A digital signal processing framework, *Proceedings of the IEEE* 75 (7) (1987) 869–891, <https://doi.org/10.1109/PROC.1987.13824>.
- [5] Y. C. Kim, E. J. Powers, Digital bispectral analysis and its applications to nonlinear wave interactions, *IEEE transactions on plasma science* 7 (2) (1979) 120–131, <https://doi.org/10.1109/TPS.1979.4317207>.
- [6] B. P. Van Milligen, C. Hidalgo, E. Sánchez, Nonlinear phenomena and intermittency in plasma turbulence, *Physical review letters* 74 (3) (1995) 395, <https://doi.org/10.1103/PhysRevLett.74.395>.
- [7] P. Diamond, M. Rosenbluth, E. Sanchez, C. Hidalgo, B. Van Milligen, T. Estrada, B. Branas, M. Hirsch, H. Hartfuss, B. Carreras, In search of the elusive zonal flow using cross-bicoherence analysis, *Physical review letters* 84 (21) (2000) 4842, <https://doi.org/10.1103/PhysRevLett.84.4842>.
- [8] N. Crocker, W. Peebles, S. Kubota, E. Fredrickson, S. Kaye, B. LeBlanc, J. Menard, Three-wave interactions between fast-ion modes in the national spherical torus experiment, *Physical review letters* 97 (4) (2006) 045002, <https://doi.org/10.1103/PhysRevLett.97.045002>.
- [9] M. Podesta, R. Bell, N. Crocker, E. Fredrickson, N. Gorelenkov, W. Heidbrink, S. Kubota, B. LeBlanc, H. Yuh, Non-linear dynamics of toroidicity-induced alfvén eigenmodes on the national spherical torus experiment, *Nuclear Fusion* 51 (6) (2011) 063035, <https://iopscience.iop.org/article/10.1088/0029-5515/51/6/063035>.
- [10] S.-I. Itoh, K. Itoh, Y. Nagashima, Y. Kosuga, On the application of cross bispectrum and cross bicoherence, *Plasma and Fusion Research* 12 (2017) 1101003–1101003, <https://doi.org/10.1585/pfr.12.1101003>.
- [11] A. Melnikov, L. Eliseev, S. Lysenko, M. Ufimtsev, V. Zenin, Study of interactions between gamas and broadband turbulence in the t-10 tokamak, *Nuclear Fusion* 57 (11) (2017) 115001, <https://doi.org/10.1088/1741-4326/aa796c>.
- [12] G. A. Riggs, S. Nogami, M. Koepke, A. Melnikov, L. Eliseev, S. Lysenko, P. Khabanov, M. Drabinskij, N. Kharchev, A. Kozachek, et al., Bispectral analysis of broadband turbulence and geodesic acoustic

- modes in the t-10 tokamak, *Journal of Plasma Physics* 87 (3) (2021) 885870301, <https://doi.org/10.1017/S0022377821000490>.
- [13] T. Seshadri, K. Subramanian, Cosmic microwave background bispectrum from primordial magnetic fields on large angular scales, *Physical review letters* 103 (8) (2009) 081303, <https://doi.org/10.1103/PhysRevLett.103.081303>.
- [14] M. Bucher, B. Van Tent, C. S. Carvalho, Detecting bispectral acoustic oscillations from inflation using a new flexible estimator, *Monthly Notices of the Royal Astronomical Society* 407 (4) (2010) 2193–2206, <https://doi.org/10.1111/j.1365-2966.2010.17089.x>.
- [15] J. Fergusson, M. Liguori, E. Shellard, General cmb and primordial bispectrum estimation: Mode expansion, map making, and measures of f nl, *Physical Review D* 82 (2) (2010) 023502, <https://doi.org/10.1103/PhysRevD.82.023502>.
- [16] N. Thyagarajan, C. L. Carilli, B. Nikolic, Detecting cosmic reionization using the bispectrum phase, *Physical Review Letters* 120 (25) (2018) 251301, <https://doi.org/10.1103/PhysRevLett.120.251301>.
- [17] B. van Tent, P. C. Delgado, R. Durrer, Constraining the bispectrum from bouncing cosmologies with planck, *Physical Review Letters* 130 (19) (2023) 191002, <https://doi.org/10.1103/PhysRevLett.130.191002>.
- [18] S. Lee, P. White, Higher-order time–frequency analysis and its application to fault detection in rotating machinery, *Mechanical Systems and Signal Processing* 11 (4) (1997) 637–650, <https://doi.org/10.1006/mssp.1997.0098>.
- [19] B. Liang, S. Iwnicki, Y. Zhao, Application of power spectrum, cepstrum, higher order spectrum and neural network analyses for induction motor fault diagnosis, *Mechanical Systems and Signal Processing* 39 (1-2) (2013) 342–360, <https://doi.org/10.1016/j.ymssp.2013.02.016>.
- [20] M. E. Iglesias Martínez, J. A. Antonino-Daviu, L. Dunai, J. A. Conejero, P. Fernández de Córdoba, Higher-order spectral analysis and artificial intelligence for diagnosing faults in electrical machines: An overview, *Mathematics* 12 (24) (2024) 4032, <https://doi.org/10.3390/math12244032>.
- [21] K. C. Chua, V. Chandran, U. R. Acharya, C. M. Lim, Application of higher order statistics/spectra in biomedical signals—A review, *Medical engineering & physics* 32 (7) (2010) 679–689, <https://doi.org/10.1016/j.medengphy.2010.04.009>.
- [22] C. M. Spooner, *Theory and application of higher-order cyclostationarity*, University of California, Davis, 1992, <https://www.proquest.com/docview/303982953>.
- [23] R. Kakarala, The bispectrum as a source of phase-sensitive invariants for fourier descriptors: a group-theoretic approach, *Journal of Mathematical Imaging and Vision* 44 (2012) 341–353, <https://doi.org/10.1007/s10851-012-0330-6>.
- [24] S. Sanborn, C. Shewmake, B. Olshausen, C. Hillar, Bispectral neural networks, *arXiv preprint arXiv:2209.03416* <https://arxiv.org/abs/2209.03416> (2022).
- [25] A. Swami, G. B. Giannakis, G. Zhou, Bibliography on higher-order statistics, *Signal processing* 60 (1) (1997) 65–126, [https://doi.org/10.1016/S0165-1684\(97\)00065-0](https://doi.org/10.1016/S0165-1684(97)00065-0).
- [26] A. Swami, J. M. Mendel, C. L. Nikias, Higher-order spectral analysis toolbox, *The Mathworks Inc* 3 (1998) 22–26.
- [27] S. A. Hosseini, M. A. Khalilzadeh, M. B. Naghibi-Sistani, V. Niazmand, Higher order spectra analysis of eeg signals in emotional stress states, in: *2010 Second international conference on information technology and computer science*, IEEE, 2010, pp. 60–63, <https://doi.org/10.1109/ITCS.2010.21>.
- [28] F. S. SAYIN, Ö. AKGÜN, Higher order spectral analysis of ventricular arrhythmic eeg signals with matlab hosa toolbox, in: *2018 6th International Conference on Control Engineering & Information Technology (CEIT)*, IEEE, 2018, pp. 1–4, <https://doi.org/10.1109/CEIT.2018.8751834>.
- [29] G. Riggs, M. Koepke, W. Heidbrink, M. Van Zeeland, D. Spong, Time-resolved biphasic signatures of quadratic nonlinearity observed in coupled alfvén eigenmodes on the diii-d tokamak, *Physics of Plasmas* 31 (4), <https://doi.org/10.1063/5.0195036> (2024).
- [30] L. Davis, J. Luxon, The doublet iii big dee project, *Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films* 3 (3) (1985) 1161–1167, <https://doi.org/10.1116/1.573097>.
- [31] W. Heidbrink, Basic physics of alfvén instabilities driven by energetic particles in toroidally confined plasmas, *Physics of Plasmas* 15 (5) (2008) 055501, <http://dx.doi.org/10.1063/1.2838239>.
- [32] L. Chen, F. Zonca, Physics of alfvén waves and energetic particles in burning plasmas, *Reviews of Modern Physics* 88 (1) (2016) 015008, <https://doi.org/10.1103/RevModPhys.88.015008>.
- [33] Y. Kim, E. Powers, Digital bispectral analysis of self-excited fluctuation spectra, *The Physics of Fluids* 21 (8) (1978) 1452–1453, <https://doi.org/10.1063/1.862365>.

- [34] S. Elgar, G. Sebert, Statistics of bicoherence and biphas, *Journal of Geophysical Research: Oceans* 94 (C8) (1989) 10993–10998, <https://doi.org/10.1029/JC094iC08p10993>.
- [35] C. L. Nikias, J. M. Mendel, Signal processing with higher-order spectra, *IEEE Signal processing magazine* 10 (3) (1993) 10–37, <https://doi.org/10.1109/79.221324>.
- [36] V. Kravtchenko-Berejnoi, F. Lefeuvre, V. Krasnosel'skikh, D. Lagoutte, On the use of tricoherent analysis to detect non-linear wave-wave interactions, *Signal processing* 42 (3) (1995) 291–309, [https://doi.org/10.1016/0165-1684\(94\)00136-N](https://doi.org/10.1016/0165-1684(94)00136-N).
- [37] S. Elgar, V. Chandran, Higher-order spectral analysis to detect nonlinear interactions in measured time series and an application to chua's circuit, *International Journal of Bifurcation and Chaos* 3 (01) (1993) 19–34, <https://doi.org/10.1142/S0218127493000039>.
- [38] W. Collis, P. White, J. Hammond, Higher-order spectra: the bispectrum and trispectrum, *Mechanical systems and signal processing* 12 (3) (1998) 375–394, <https://doi.org/10.1006/mssp.1997.0145>.
- [39] E. Strait, Magnetic diagnostic system of the diii-d tokamak, *Review of scientific instruments* 77 (2) (2006) 023502, <https://doi.org/10.1063/1.2166493>.
- [40] A. Papoulis, *The Fourier integral and its applications*, McGraw-Hill New York, 1967.
- [41] J. Allen, Short term spectral analysis, synthesis, and modification by discrete fourier transform, *IEEE Transactions on Acoustics, Speech, and Signal Processing* 25 (3) (1977) 235–238, <https://doi.org/10.1109/TASSP.1977.1162950>.
- [42] J. Van den Berg, *Wavelets in physics*, Wavelets in Physics (2004).
- [43] J. Morlet, Sampling theory and wave propagation, in: *Issues in acoustic Signal image processing and recognition*, Springer, 1983, pp. 233–261, https://doi.org/10.1007/978-3-642-82002-1_12.
- [44] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, T. E. Oliphant, *Array programming with NumPy*, *Nature* 585 (7825) (2020) 357–362. doi:10.1038/s41586-020-2649-2. URL <https://doi.org/10.1038/s41586-020-2649-2>
- [45] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, *Nature Methods* 17 (2020) 261–272. doi:10.1038/s41592-019-0686-2.
- [46] J. D. Hunter, Matplotlib: A 2d graphics environment, *Computing in Science & Engineering* 9 (3) (2007) 90–95. doi:10.1109/MCSE.2007.55.
- [47] F. Lundh, An introduction to tkinter, URL: www.pythonware.com/library/tkinter/introduction/index.htm (1999).
- [48] G. A. Riggs, *Interpretations of bicoherence in space & lab plasma dynamics*, West Virginia University, 2020, <https://www.proquest.com/docview/2599117379>.
- [49] R. Stauber, *Applicability of bispectral analysis to unstable plasma waves*, West Virginia University, 1995.
- [50] F. J. Harris, On the use of windows for harmonic analysis with the discrete fourier transform, *Proceedings of the IEEE* 66 (1) (1978) 51–83, <https://doi.org/10.1109/PROC.1978.10837>.
- [51] L. Salvatore, A. Trotta, Flat-top windows for pwm waveform processing via dft, in: *IEE Proceedings B (Electric Power Applications)*, Vol. 135, IET, 1988, pp. 346–361, <https://doi.org/10.1049/ip-b.1988.0038>.
- [52] B. P. Van Milligen, E. Sanchez, T. Estrada, C. Hidalgo, B. Brañas, B. Carreras, L. García, Wavelet bicoherence: a new turbulence analysis tool, *Physics of Plasmas* 2 (8) (1995) 3017–3032, <https://doi.org/10.1063/1.871199>.
- [53] J. A. Schulte, Wavelet analysis for non-stationary, nonlinear time series, *Nonlinear Processes in Geophysics* 23 (4) (2016) 257–267, <https://doi.org/10.5194/npg-23-257-2016>.
- [54] S. H. Jacobson, E. Yücesan, Analyzing the performance of generalized hill climbing algorithms, *Journal of Heuristics* 10 (4) (2004) 387–405, <https://doi.org/10.1023/B:HEUR.0000034712.48917.a9>.
- [55] V. Chandran, On the computation and interpretation of auto-and cross-trispectra, in: *Proceedings of ICASSP'94. IEEE International Conference on Acoustics, Speech and Signal Processing*, Vol. 4, IEEE, 1994, pp. IV–445, <https://doi.org/10.1109/ICASSP.1994.389784>.

- [56] J. Fackrell, S. McLaughlin, P. White, Practical issues concerning the use of the bicoherence for the detection of quadratic phase coupling, in: Proceedings of the IEEE Workshop on HOS, Citeseer, 1995, pp. 310–314, <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=8c456ca10994ed212d596fe1f7be397f82b4c2f2>.
- [57] P. Z. Poloskei, G. Papp, L. Horvath, G. I. Pokol, et al., Bicoherence analysis of nonstationary and nonlinear processes, arXiv preprint arXiv:1811.02973<https://arxiv.org/abs/1811.02973> (2018).
- [58] V. Chandran, S. Elgar, B. Vanhoff, Statistics of tricoherence, IEEE transactions on signal processing 42 (12) (1994) 3430–3440, <https://doi.org/10.1109/78.340777>.
- [59] B. Efron, An introduction to the bootstrap, Chapman and Hall/CRC, 1994.
- [60] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: a practical and powerful approach to multiple testing, Journal of the Royal statistical society: series B (Methodological) 57 (1) (1995) 289–300, <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- [61] B. Babadi, E. N. Brown, A review of multitaper spectral analysis, IEEE Transactions on Biomedical Engineering 61 (5) (2014) 1555–1564, <https://doi.org/10.1109/TBME.2014.2311996>.
- [62] H. Von Storch, A. Navarra, Analysis of climate variability: applications of statistical techniques, Springer Science & Business Media, 1999.
- [63] G. Riggs, M. Koepke, W. Heidbrink, M. Van Zeeland, D. Spong, Instantaneous difference-frequency locking observed in coupled eigenmodes on the diii-d tokamak, submitted to *Nuclear Fusion* (2026).